

Making OpenAD Robust to Query Phrasing

in 3D Affordance Detection

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Problem & contribution

Affordance detection: *where* on a 3D object can I act (grasp a handle, sit on a seat, cut with a blade)?

OpenAD handles this as an open-vocabulary task. It matches each point's PointNet++ feature to a query embedding from *frozen CLIP*. However, it is trained using only *single-word* queries.

Users may not say “grasp”—they might ask “Where can I grasp this object?” How well does the model handle that change in wording?

Our approach (pretrained weights, no new architecture):

- reproduce OpenAD and measure its sensitivity to phrasing;
- improve its robustness with lightweight prompt-augmented fine-tuning;
- provide an interactive demonstration.

Headline

Rephrasing costs **~12 mIoU**. Fine-tuning **101k params (5.7%)** improves held-out questions by **11.2** and descriptions by **10.5 mIoU**.

3D AffordanceNet (full-shape, as shipped with OpenAD)

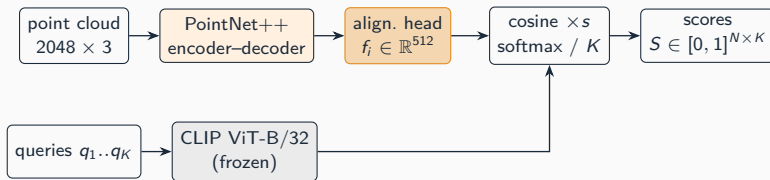
- Each object has 2,048 points and one of **19 classes**: 18 affordances plus none.
- The validation split contains **2,285 objects** and is highly imbalanced. Table and Chair make up 62% of it.

Metrics. We report per-point **mIoU** over 19 classes, point accuracy, and mean class accuracy. Our implementation matches OpenAD's evaluator and is covered by unit tests.

Phrasing conditions. We study five affordances: grasp, contain, sittable, pourable, and cut. Each appears as a label, question, or description. The fine-tuning and held-out phrase sets were **fixed before training**, and a test checks that they do not overlap.

Evaluation noise. Because farthest-point sampling starts randomly, repeated runs vary by about **0.4 mIoU**. We treat differences below 0.5 as equivalent.

Method: architecture + our modification



■ trainable (head + last prop. layer, 101k) ■ frozen backbone ■ frozen CLIP

Unchanged from OpenAD

- cosine align $\times$ learnable scale s , softmax
- class-weighted NLL (cube-root inv-freq)
- Adam

Our modification: prompt-augmented fine-tuning

- At each step, sample one paraphrase for every studied affordance. Labels and loss remain unchanged.
- Train only the **head and fp1** (5.7% of the network) for 8 epochs at LR 10^{-4} .
- Select the epoch using canonical validation mIoU, without using held-out phrases.

Models compared

- majority-class and random baselines
- pretrained OpenAD from a pinned commit
- our fine-tuned models at three capacities

Capacity ladder (same data/seed):

head 67k $\rightarrow$ head+fp1 101k $\rightarrow$ decoder 743k.

Fair comparison

- Both models are evaluated in the same script run.
- They use the same objects, order, metrics, and query lists.
- **Only the model weights differ** in the model comparison.

Implementation

- PyTorch 2.5 / CUDA 12.1, 1 $\times$ RTX 4060
- 16 GPU-free unit tests

Reproduction validates the harness

Setting	Method	mIoU	Acc	mAcc
Closed-set	OpenAD (cited)	42.00	69.03	67.31
	ours	41.25	68.51	66.73
Open-vocab	OpenAD (cited)	14.37	46.31	19.51
	ours	14.40	46.33	19.63
Floor	majority (none)	2.27	43.04	5.26
	random	1.47	5.28	5.43

Our open-vocabulary score of **14.40** closely matches the reported 14.37. The majority baseline reaches 43% point accuracy but only 2.27 mIoU, showing why mIoU is more informative for this imbalanced dataset.

The problem: phrasing breaks the pretrained model

Affordance	Canon.	Ques.	Descr.
grasp	42.6	0.0	0.4
contain	43.9	0.0	0.3
sittable	59.6	2.0	21.8
pourable	41.5	0.0	0.0
cut	55.2	43.2	37.1
mIoU (19)	41.5	29.6	30.3

Same intent, different wording

Canonical grasp

Question "Where should I take hold of this?"

Description "the surface meant for gripping by hand"

Rephrasing the queries as questions or descriptions reduces mIoU by about **12 points**.

Grasp, contain, and pourable fall close to zero, while *cut* remains relatively strong.

Useful geometry, weak text alignment

Across points: rephrased grasp scores handle 0.30 > body 0.10 $\Rightarrow$ it still locates the handle region.

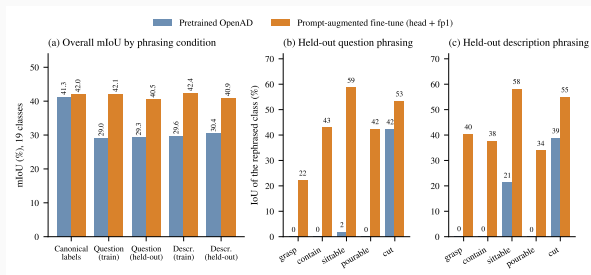
At the handle: grasp 0.30 < lift 0.40 $\Rightarrow$ another query wins the final class label.

PointNet++ still distinguishes the useful region; its match to the rephrased text is too weak.

Results: fine-tuning improves held-out phrasing

Condition	Pre	FT	Δ
Canonical	41.3	42.0	+0.7
Question, seen	29.0	42.1	+13.1
Question, held-out	29.3	40.5	+11.2
Description, seen	29.6	42.4	+12.7
Descr., held-out	30.4	40.9	+10.5

- Held-out questions improve by **11.2 mIoU**; descriptions improve by **10.5**.
- Held-out scores are close to seen scores, suggesting generalisation rather than exact phrase memorisation.
- Preserves canonical-word performance (+0.7).



Ablation: how much of the network to train?

Model	Params	Canon.	Q held-out	D held-out
Head only	67,073	+0.49	+10.53	+10.04
Head+fp1 (reported)	101,377	+0.74	+11.21	+10.48
Decoder (fp1–fp3)	743,041	+0.54	+10.91	+10.28

Values show the mIoU improvement after fine-tuning.
Head alone already gives +10.5/ +10.0.

Performance stays nearly constant across an **11× parameter range**. Most of the improvement therefore comes from the alignment head, so the encoder can remain frozen.

Training behaviour & open-vocabulary transfer

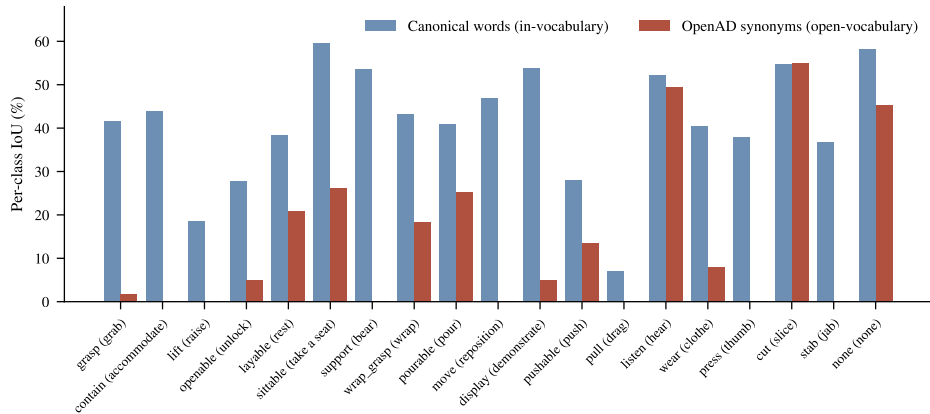
Epoch	Loss	Val mIoU
1	0.701	41.72
4	0.644	42.46
8	0.639	42.15

Validation mIoU peaks at epoch 4 and remains stable afterwards.

Synonym	Pre	FT	Δ
grasp (grab)	1.7	36.3	+34.6
sittable (take a seat)	26.2	57.9	+31.7
contain (accommodate)	0.2	0.0	-0.2
13 unstudied (mean)	9.22	9.17	-0.05
mIoU (19)	14.40	18.66	+4.26

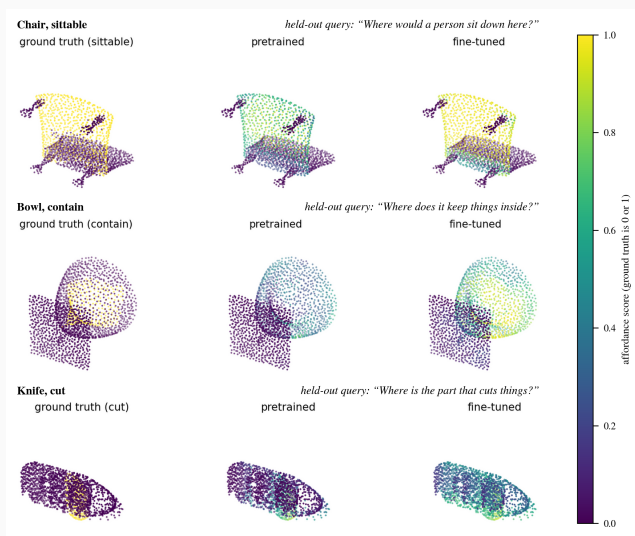
The five fine-tuned affordances account for 97% of the gain. The other classes remain stable.

Per-class IoU: canonical vs. synonym



Ten of 18 synonyms fall below 5% IoU; two (slice, hear) nearly match the canonical word. Transfer is therefore **non-uniform** and depends on the chosen synonym.

Qualitative, failure cases & demo



Successes: The chair and bowl regions are recovered, and the knife has a sharper peak.

Failure: Neither model detects the grasp region of the bag. The split contains only 12 bags, and their handles vary considerably.

Demo: The Gradio and Plotly interface lets users choose an object and model, enter a query, inspect a rotatable heat map, compare phrasings, and identify low-confidence results.

Conclusion

- We reproduced OpenAD (**14.40** vs. 14.37). Natural rephrasing of five affordances reduces performance by about **12 mIoU**.
- **Prompt-augmented fine-tuning** of 101k params (5.7%) improves held-out questions by 11.2 and descriptions by 10.5 mIoU, while preserving canonical performance.
- The alignment head provides most of the gain across an $11\times$ parameter range, indicating that the main mismatch is in **point-to-text alignment** rather than geometry.

Limitations and next steps: five affordances, ten held-out phrases, and one seed. Future work should test more prompts and seeds, evaluate synonyms at every depth, and explore objectives that encourage consistent predictions across paraphrases.

Thank you. Questions?

github.com/mehreen019/Open-Vocabulary-3D-Affordance-Detection